

Titanic Dataset Data Preprocessing and Feature Engineering Report

Project Title

Data Preprocessing, Feature Engineering, and Preparation of the Titanic Dataset for Machine Learning

1. Introduction

Data preprocessing is one of the most important phases of every Machine Learning project. Raw datasets usually contain missing values, duplicate records, inconsistent formats, and categorical data that cannot be directly processed by machine learning algorithms.

The objective of this project is to clean, transform, and prepare the Titanic dataset so that it becomes suitable for future predictive modeling. The notebook follows a complete preprocessing pipeline using Python and Scikit-learn.

2. Project Objectives

The main objectives of this project are:

- Load and understand the Titanic dataset.
 - Explore dataset characteristics.
 - Detect and remove duplicate records.
 - Handle missing values.
 - Perform feature engineering.
 - Encode categorical variables.
 - Scale numerical features.
 - Detect outliers.
 - Split the dataset into training and testing sets.
 - Build an automated preprocessing pipeline.
 - Validate the processed dataset before machine learning.
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3. Technologies Used

Technology	Purpose
Python	Programming language

Technology	Purpose
Pandas	Data manipulation
NumPy	Numerical computations
Seaborn	Dataset loading and visualization
Matplotlib	Graph plotting
Scikit-learn	Machine Learning preprocessing

4. Libraries Used

4.1 Pandas

Used for:

- Loading dataset
- Cleaning data
- Manipulating rows and columns
- Handling missing values

Example:

```
import pandas as pd
```

4.2 NumPy

Used for mathematical operations and numerical processing.

```
import numpy as np
```

4.3 Seaborn

Used to load the Titanic dataset and visualize outliers.

```
import seaborn as sns
```

Dataset:

```
df = sns.load_dataset('titanic')
```

4.4 Matplotlib

Used to visualize data using boxplots.

```
import matplotlib.pyplot as plt
```

4.5 Scikit-learn

Several preprocessing tools are used.

LabelEncoder

Converts categorical labels into numerical values.

```
from sklearn.preprocessing import LabelEncoder
```

StandardScaler

Standardizes numerical features.

```
from sklearn.preprocessing import StandardScaler
```

train_test_split

Splits the dataset into training and testing datasets.

```
from sklearn.model_selection import train_test_split
```

Pipeline

Creates an automated preprocessing workflow.

```
from sklearn.pipeline import Pipeline
```

ColumnTransformer

Applies different preprocessing methods to numerical and categorical columns.

```
from sklearn.compose import ColumnTransformer
```

SimpleImputer

Fills missing values.

```
from sklearn.impute import SimpleImputer
```

OneHotEncoder

Converts categorical variables into binary vectors.

```
from sklearn.preprocessing import OneHotEncoder
```

5. Dataset Description

Dataset Name:

Titanic Dataset

Source:

Seaborn Library

Prediction Target:

Survived

Each passenger contains information such as:

- Passenger Class
 - Sex
 - Age
 - Fare
 - Embarked Port
 - Number of siblings/spouses
 - Number of parents/children
 - Survival Status
-

6. Data Exploration

The notebook begins with exploratory data analysis.

The following operations are performed:

- Display first five records
- Check dataset dimensions
- Display column data types
- Generate descriptive statistics
- Identify missing values
- Count duplicate rows

These steps help understand the quality and structure of the dataset before preprocessing.

7. Data Cleaning

The following preprocessing tasks are completed.

Duplicate Removal

Duplicate records are removed using:

```
drop_duplicates()
```

Missing Value Handling

Different strategies are used depending on the column.

Age

Missing values replaced using the median.

Reason:

Median is less sensitive to outliers.

Fare

Missing values replaced using the median.

Embarked

Missing values replaced using the most frequent category (mode).

Deck

The Deck column is removed because it contains a large number of missing values.

8. Feature Engineering

Feature engineering creates new useful variables.

Family Size

A new feature called **FamilySize** is created.

Formula:

```
FamilySize = SibSp + Parch + 1
```

This represents the total number of family members travelling together.

Alone Feature

The notebook attempts to create an **Alone** feature for passengers travelling alone. However, the current implementation uses the comparison operator (==) instead of assignment (=), so the values are not actually stored. This should be corrected if the feature is intended to be used.

9. Data Transformation

Encoding

Categorical variables cannot be processed directly by most machine learning algorithms.

The following encoding is performed.

Sex

Male → 0

Female → 1

Embarked

Encoded using LabelEncoder.

Title

The notebook checks for a Title column.

If available, it is encoded.

10. Feature Scaling

The following numerical columns are standardized.

- Age
- Fare
- FamilySize

Scaling ensures every feature has a similar range, preventing larger-valued variables from dominating the learning process.

11. Outlier Detection

A boxplot is created for the Age column.

Purpose:

- Detect extreme values
 - Understand distribution
 - Decide whether outlier treatment is required
-

12. Dataset Preparation

The dataset is divided into:

Features (X)

Independent variables.

Target (y)

Survived

Train-Test Split

The notebook uses:

- 80% Training
- 20% Testing

Parameters:

- random_state = 42
- stratify = y

Stratification preserves the class distribution in both training and testing datasets.

13. Preprocessing Pipeline

A Scikit-learn preprocessing pipeline is constructed.

Numerical Pipeline

Steps:

1. Median Imputation
 2. Standard Scaling
-

Categorical Pipeline

Steps:

1. Most Frequent Imputation
 2. One-Hot Encoding
-

ColumnTransformer

Different preprocessing techniques are automatically applied to:

Numerical Features

- Age
- Fare
- SibSp
- Parch

Categorical Features

- Sex
- Embarked
- Pclass

This modular pipeline makes preprocessing reproducible and reduces the risk of errors.

14. Validation

Several validation checks are performed.

Missing Values

Verified that no missing values remain after preprocessing.

Data Types

Ensured all transformed features are numerical.

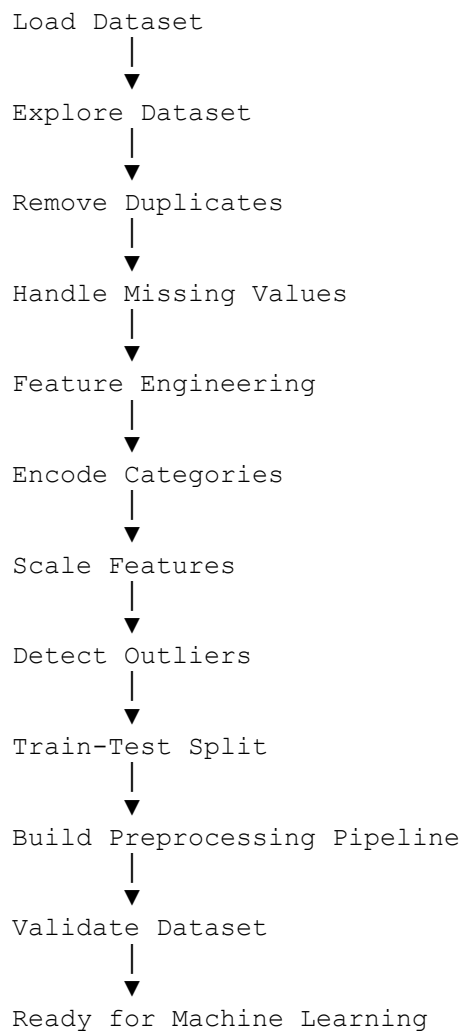
Data Leakage

The preprocessing pipeline is fitted only on the training data and then applied to the test data, helping to prevent data leakage.

Class Distribution

The distribution of the target variable is compared across the original, training, and testing datasets to confirm stratification has preserved class balance.

15. Project Workflow



16. Results

The notebook successfully:

- Loaded the Titanic dataset.
- Explored dataset characteristics.
- Removed duplicate records.
- Handled missing values.
- Created new features.
- Encoded categorical variables.
- Standardized numerical variables.
- Prepared training and testing datasets.
- Built an automated preprocessing pipeline.
- Validated the processed data.

17. Conclusion

This project demonstrates a complete and systematic data preprocessing workflow using Python and Scikit-learn. The Titanic dataset is transformed from raw data into a clean, standardized, and machine-learning-ready format through data cleaning, feature engineering, encoding, scaling, and pipeline construction. The resulting dataset provides a reliable foundation for developing predictive models while following best practices to ensure reproducibility and reduce the risk of data leakage.